Operating-system structures exercises

State the model, intermediate state, and reasoning before checking the separate solutions.

OS02-E01 API and system call

A library function send_record() may copy data to a kernel buffer and enqueue device work.

1. Which steps necessarily execute in kernel mode?
2. Does library success imply durable data?
3. What must the kernel validate when copying a user pointer?

OS02-E02 Microkernel trade-offs

A claim says moving every filesystem/network service outside the kernel makes the system safer.

1. Explain the supporting isolation argument.
2. Identify at least two costs or new risks.
3. Give a mitigation that does not assume architecture alone establishes safety.

OS02-E03 Boot trust chain

Firmware loads an unsigned loader, which loads a signed kernel.

1. Can the kernel signature stop this loader replacing the kernel?
2. Where must verification occur for a verifiable chain?
3. What controls must accompany module signatures?

OS02-E04 Short writes and copy invariant

Fixed source 6000 B, empty destination, buffer 4096 B, no other writer. Reads return 4096,1904,0; positive writes return 1000,3096,1024,880.

1. Trace both offsets and total read/write bytes.
2. A faulty variant performs only one write per read. Its first writes for the two chunks return 1000 and1024. How many bytes are lost? Do not reuse3096, which belonged to a different remaining-range request, as a return from the new1904 B request.
3. State a general invariant and handle short writes/EINTR.
4. Does destination fsync alone establish a durable new name and atomic replacement?

Operating-system structures exercises (continued)

OS02-E05 Pipeline references and EOF

Pipe gives fd3 read/fd4 write. Shell forks cat and wc. Cat dup2s4 to stdout; wc dup2s3 to stdin.

1. List post-exec pipe references each process should retain.
2. Cat writes6000 B and exits. Give a concrete reason wc still cannot observe EOF.
3. Can cat fail opening nonexistent a while wc outputs0 and exits0?
4. Do two forks and two execs create four children?

OS02-E06 ABI range and TOCTOU

A model ABI copies24 B arguments and an8 B pathname. size_t is32-bit, user upper bound U=0xC0000000, ranges are nonempty, and elements occupy4 B.

1. Compute copied bytes and distinguish API/ABI/implementation.
2. For p=0xBFFFFF00, test byte lengths0x100 and0x101. Give an overflow-safe condition.
3. Compute mathematical and truncated values of0x40000001×4.
4. Why is checking a VMA before direct dereference insufficient?

OS02-E07 Policy and layer dependency

One CPU, A/B/C bursts5/3/1 ms, all ready0, RR order A, B, C, no I/O/switch cost. Separately, pager→driver→scheduler→pager.

1. Compute completion/response for q 1 and q 4.
2. Compute mean response and actual task-switch counts.
3. Identify mechanisms and policy.
4. Why is the cycle incompatible with strict layers, and how can it be removed concretely?

OS02-E08 Trace, tail latency, and race

Request starts0, blocks1, ready8, runs9, returns 10. Two threads separately increment x=0 non-atomically. Latencies are1,1,1,1,1,1,1,100 ms.

1. Split10 ms into CPU, blocked, and ready wait.
2. Compute mean and nearest-rank p99.
3. Give a lost update and explain printf's possible masking effect.
4. Does a mutex around only each write repair the race?

Operating-system structures exercises (continued)

OS02-E09 Root closure and trust chain

Root uses storage S/filesystem F. Other requirements are built in. Modules are accessible only from initramfs or mounted root. A separate firmware→loader→kernel signature policy applies.

1. Can root mount if S is root-only and F built-in?
2. Why do S/F in initramfs still not prove successful boot?
3. Is a kernel signature checked solely by an untrusted loader sufficient?
4. Distinguish digest measurement, valid signature, and anti-rollback.

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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https://github.com/xinhi6716/computer-science-self-study

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